Proper Besag model for spatial effects

Parametrization

The proper version of the Besag model for random vector $\mathbf{x} = (x_1, \ldots, x_n)$ is defined as

$$x_i | x_{-i}, \tau, \phi \quad \sim \quad \mathcal{N}\left(\frac{\phi}{1 + \phi n_i} \sum_{i \sim j} x_j, \frac{1}{(1 + \phi n_i)\tau} \right) \tag{1}$$

where n_i is the number of neighbours of node i, $i \sim j$ indicates that the two nodes i and j are neighbours, $\phi > 0$ is as weight parameter and $\tau > 0$ is a "precision-like" (or scaling) parameter.

Hyperparameters

The precision parameter τ is represented as

$$\theta_1 = \log \tau$$

and the prior is defined on θ_1.

The weight parameter ϕ is represented as

$$\theta_2 = \log \phi$$

and the prior is defined on θ_2.

Specification

The besag model is specified inside the `f()` function as

```
f(<whatever>,model="besagproper",graph=<graph>,
  hyper=<hyper>)
```

The neighbourhood structure of **x** is passed to the program through the `graph` argument. The structure of this file is described below.

Hyperparameter spesification and default values

doc `A proper version of the Besag model`

hyper

- **theta1**
  - **hyperid** `12001`
  - **name** `log precision`
  - **short.name** `prec`
  - **prior** `loggamma`
  - **param** `1 5e-04`
  - **initial** `2`
  - **fixed** `FALSE`
  - **to.theta** `function(x) log(x)`
  - **from.theta** `function(x) exp(x)`
- **theta2**
  - **hyperid** `12002`

name `log diagonal`
short.name `diag`
prior `loggamma`
param `1 1`
initial `1`
fixed `FALSE`
to.theta `function(x) log(x)`
from.theta `function(x) exp(x)`

constr `FALSE`

nrow.ncol `FALSE`

augmented `FALSE`

aug.factor `1`

aug.constr

n.div.by

n.required `TRUE`

set.default.values `TRUE`

pdf `besagproper`

Structure of the graph file

OLD: We describe the required format for the graph file using a small example. Let the file `gra.dat`, relative to a small graph of only 5 elements, be

```
5
1  1  2
2  2  1  3
3  3  2  4  5
4  1  3
5  1  3
```

Line 1 declares the total number of nodes in the graph (5), then, in lines 2-6 each node is described. For example, line 4 states that node 3 has 4 neighbours and these are nodes 2, 4 and 5.

The graph file can either have nodes indexed from 1 to n, or from 0 to $n-1$. Note that in the latter case, node i seen from R corresponds to node $i-1$ in the 0-indexed graph.

Example

To be added

Notes

Add notes later ?